

DP Memorization Study: Experiment Matrix

Fixed setup

- **Model:** Mistral-7B-Instruct-v0.3. **Template:** `Diagnosis: {disease}. {answer}`.
- **Dataset:** https://huggingface.co/datasets/sarus-tech/medical_v3. 99 diseases (classes), highly unbalanced and long tailed.
- We are not doing classification of the diseases, but next token prediction.
- **DP:** DP-SGD, Poisson sampling rate `q`. **Fixed-target- ϵ accounting:** fix step budget `s` up front, accountant derives σ for the target ϵ . $\delta = \min(1/N, 1-e5)$.
- **Optimizer:** AdamW, **single global LR** (HPO deferred to later stages), fixed batch & seq len.
- **ϵ grid (DP):** {1, 4, 16, 64}. ∞ (non-private) = **memorization ceiling**
- **Clip norm default:** Default 1. C only varies in E3.
- **Batch size default:** 1024
- **Learning rate default:** 0.00026
- **LoRA rank default:** 8
- **Seeds:** 3 per cell (BCa bootstrap CIs).
- **Splits (load-bearing):**
 - *Held-out patients of seen diseases:* concept generalization.
 - *Held-out diseases (never trained):* negative control, accuracy must stay at chance.
 - *Matched canary sets* `c_in` / `c_out` $\rightarrow$ leakage in-vs-out.

Experiments

Runs	Exp	Sweeps	Fixed	What it tests	How / readout
5	E1 $\rightarrow$ Privacy sweep (headline gap)	$\epsilon \in \{\infty, 64, 16, 4, 1\}$	default C, rank, learning rate, batch size	Does record memorization collapse at <i>higher</i> ϵ than concept learning?	On the <i>same</i> models: concept (held-out-patient disease acc + correct-disease log-prob) and leakage (μ -GDP, reconstruction lift, PII recon). Plot both axes vs $\epsilon \rightarrow$ the gap.
	E2 $\rightarrow$ Concept phase diagram	disease frequency bin $\times \epsilon \in \{\infty, 16, 4, 1\}$ (reuses E1 checkpoints, no new training)	default C, rank, learning rate, batch size	Minimum frequency to learn a concept	Stratify diseases by natural frequency; held-out-patient concept acc per bin $\times \epsilon \rightarrow$ 2D phase diagram.
	E6 $\rightarrow$ Similar-	similarity clusters	default C, rank,	Does DP raise	Confusion

	disease interference	$\times \epsilon \in \{\infty, 16, 4, 1\}$ (reuses E1 checkpoints, no new training)	learning rate, batch size	confusion between similar diseases; rare→common absorption?	matrices among cluster members vs ϵ .
4	E3 → Clipping & long-tail	$C \in \{1, 10, 50, 500\}$ (span grad- norm tail; confirm vs pilot percentiles)	$\epsilon = 4$, rank 16	Does clipping suppress rare/atypical diseases more?	Record clipped gradients during training Per-frequency- bin concept acc + per-step clipped- fraction logged by frequency bin.
4	E4 → Rank	rank $\in$ $\{4, 8, 16, 32\}$	$\epsilon \in \{\infty, 4\}$, C default	Contested rank and memorization	μ -GDP + reconstruction lift on Tier-2 canaries.
	E5 → Capability retention	(reuses E1 checkpoints, no new training)		Forgetting vs ϵ regularization (less forgetting) vs degradation.	Benchmark suite (MMLU subset, ARC, instruction set) + forgetting = $\text{score}(\theta_{\text{pt}}) - \text{score}(\theta_{\text{ft}})$ vs ϵ .

MI score: $s(c) = \log \Pr(c|\theta_{\text{ft}}) - \log \Pr(c|\theta_{\text{pt}})$.

Reconstruction: 50-tok prefix → greedy 10-tok suffix → lift over C_{out} at edit distance $d \in \{0..5\}$.